BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

PHYSICS — SSC-I (9th Grade)

Syllabus: Unit 5 — Pressure and Deformation in Solids

ANSWER KEY (Version 1)

Total Marks: 65 | **Time Allowed: 2 hours 30 minutes** | **Version: 1**

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	Answer 1 mk	Explanation
1	The ability of a deformed body to return to its original shape when deforming force is removed is called:	**B** Elasticity	**Elasticity** is defined as the ability of a deformed body to return to its original shape. **Plasticity** is the opposite — materials that remain deformed. **Rigidity** means resistance to deformation altogether.
2	Hooke's law holds good up to the:	**C** Proportional limit	Hooke's Law ($F \propto x$) holds only within the **limit of proportionality** (point A on stress-strain curve). Beyond this, the graph curves. The **elastic limit** is slightly beyond point A; beyond it, permanent deformation occurs.
3	The SI unit of the spring constant is:	**A** N/m	From $k = F/x$, units are **newtons per metre (N/m)**. A stiffer spring has a larger k value. N·m is the unit of torque/energy, not spring constant.
4	The SI unit of pressure is:	**D** pascal	The SI unit of pressure is the **pascal (Pa)**, equal to 1 N/m^2. The *bar* and *psi* are non-SI units; newton is the unit of force, not pressure.
5	A 2 kg mass hung on a spring displaces it 5 cm. Spring constant is:	**A** 400 N/m	$F = mg = 2 \times 10 =$ **20 N**; $x = 5$ cm $= 0.05$ m; $k = F/x = 20/0.05 =$ **400 N/m**. Common error: forgetting to convert cm to m, giving 40 instead.
6	The most elastic material among the following is:	**D** Steel	**Steel** has the highest **Young's modulus** (elasticity). Though rubber stretches a lot, it is not the most elastic in the physics sense — the most elastic material returns closest to its original shape under stress. The textbook (exercise Q1) confirms steel.
7	Atmospheric pressure at sea level is approximately:	**B** 1.013×10^5 Pa	1 atm = $\mathbf{1.013 \times 10^5}$ **Pa** = 101,325 Pa ≈ 101.3 kPa. Option A (10^3 Pa) is only ~1% of the correct value; the other options are not standard values.
8	Pressure in a liquid at depth h is given by:	**C** $P = \rho gh$	Derived from $F = W = mg = \rho Ahg$ and $P = F/A$, giving $\mathbf{P = \rho gh}$. It depends on **density** ρ, **gravity** g, and **depth** h — not on the shape or volume of the container.
9	Atmospheric pressure is commonly measured using:	**B** Barometer	A **barometer** measures atmospheric pressure. A **manometer** measures gauge pressure of a fluid relative to atmospheric. A hygrometer measures humidity; a thermometer measures temperature.
10	Pascal's principle states that an external pressure applied to an enclosed fluid is transmitted:	**C** Unchanged to every point	**Pascal's Principle**: "An external pressure applied to an enclosed fluid is transmitted unchanged to every point within the fluid." This is the basis of hydraulic systems — pressure doesn't diminish or increase as it spreads.
	The atmospheric pressure will be smallest at:	**D**	Atmospheric pressure **decreases with altitude**. **Murree** is the highest location among the options (~2300 m above

11		Murree	sea level). Karachi is at sea level; Islamabad and Lahore are on the plains.
12	In a hydraulic lift, if piston 2 area is 100 times piston 1 area, the output force is:	**A** 100 times input force	From $F_2 = (A_2/A_1) \times F_1$, if $A_2/A_1 = 100$, then $F_2 = \mathbf{100\ F_1}$. This force multiplication is why hydraulic lifts can raise cars with relatively small input forces.

SECTION B — Short Answer Questions (11 × 3 = 33 Marks)

2(i) Elasticity and Elastic Limit 3 marks

Elasticity 1 mk: The ability of a deformed body to return to its original shape and size when the deforming force is removed. Examples of elastic materials: spring, rubber band.

Elastic Limit 1 mk: The maximum deforming force beyond which a material will not return to its original dimensions when the force is removed. Beyond this limit, permanent deformation occurs.

Inelastic material example 1 mk: Plasticine, clay, dough — these remain deformed after the force is removed.

OR

Inelastic materials 1 mk: Materials that do *not* return to their original shape after the deforming force is removed. They undergo **permanent deformation**.

Three examples 1 mk: (1) Plasticine, (2) Clay, (3) Dough.

Reason 1 mk: In inelastic materials, the internal atomic/molecular bonds are broken or rearranged permanently when stretched beyond their elastic limit, so there is no restoring force to bring them back.

2(ii) Hooke's Law 3 marks

Statement 1 mk: "Within the elastic limits, the extension (or compression) of a spring is directly proportional to the applied force."

Mathematical form 1 mk: $F_{res} = -kx$ or $k = F/x$

Spring constant k 1 mk: The ratio of the restoring force to the extension. It represents the stiffness of the spring. SI unit: $\mathbf{N\ m^{-1}}$ (newtons per metre).

OR

Force-Extension Graph:

```
  Force (F)
       |        * Breaking point (D)
       |      *
       |    * C (Ultimate stress)
       |  * B (Elastic limit)
       |* A (Proportional limit / Limit of proportionality)
       |/
       +-------------------------> Extension (x)
       O
```

1 mk Straight line from O to A (Hooke's Law obeyed — F directly proportional to x).

1 mk Point A = limit of proportionality. The gradient of the straight section = spring constant k.

1 mk Beyond A: graph curves; beyond B (elastic limit): permanent deformation; at D: material breaks.

2(iii) Spring Constant Calculation `3 marks`

Given: mass m = 60 kg; compression x = 8 cm = 0.08 m; g = 10 m/s²

Force `1 mk`: F = mg = 60 × 10 = **600 N**

Formula & substitution `1 mk`: `k = F/x = 600 N / 0.08 m`

Answer `1 mk`: `k = 7,500 N/m`

OR

Three applications of Hooke's Law:

- **Balance Wheel of Mechanical Watches** `1 mk`: A spiral torsion (hair) spring returns the balance wheel to its centre, keeping time constant. The elasticity of the spring regulates the oscillation period.
- **Spring Scale (Spring Balance)** `1 mk`: The extension of the spring is directly proportional to the weight hung on it (Hooke's Law). A gauge converts extension to a weight reading.
- **Galvanometer** `1 mk`: A hair spring restores the pointer and makes the deflection proportional to the current. Since force ∝ current (by Hooke's Law), an analogue scale can measure current directly.

2(iv) Pressure: Definition + Sharp Knife `3 marks`

Definition `1 mk`: **Pressure** is defined as force applied per unit area. Formula: `P = F/A`. SI unit: **pascal (Pa)** = 1 N/m².

Effect of area `1 mk`: For the same force, a smaller area produces higher pressure. A **sharp knife** has a very small cutting edge area, concentrating the same cutting force into tiny area, creating very high pressure that easily cuts through material.

Blunt knife comparison `1 mk`: A **blunt knife** has a larger edge area, so the same force is spread over more area, resulting in lower pressure — it cannot cut as easily.

OR

Given: m = 50 kg; A = 0.02 m²; g = 9.8 m/s²

Weight `1 mk`: F = W = mg = 50 × 9.8 = **490 N**

Formula `1 mk`: `P = F/A = 490 N / 0.02 m²`

Answer `1 mk`: `P = 24,500 Pa = 24.5 kPa`

2(v) Atmospheric Pressure `3 marks`

Definition `1 mk`: "The pressure exerted by the weight of atmospheric particles on the surface of the earth and all objects on it is called **atmospheric pressure**." Value at sea level: 1.013×10^5 Pa = 1 atm.

Why we don't feel it `2 mk`: We do not feel the atmospheric pressure because the pressure inside our bodies (in blood, lungs, etc.) equals the atmospheric pressure outside. There is **equal pressure in all directions** — our bodies are in equilibrium with the atmosphere. The net force on any surface of the body is zero because pressure acts inward and outward equally.

OR

Drinking through a straw `1.5 mk`: Sucking lowers the **air pressure** in the mouth below atmospheric pressure. The atmospheric pressure acting on the surface of the liquid in the glass is now *greater* than the pressure in the mouth. This pressure difference **pushes the liquid up the straw** into the mouth.

Drawing liquid by syringe `1.5 mk`: Pulling the piston back **decreases pressure** inside the cylinder. Atmospheric pressure acting on the liquid surface outside then drives the liquid into the syringe through the needle.

2(vi) Mercury Barometer 3 marks

Construction 1 mk: A long glass tube (about 80 cm) closed at one end is filled with **mercury** and inverted into a container (trough) of mercury. The closed end creates a vacuum above the mercury column.

Working 1 mk: The mercury in the tube falls until the **weight of the mercury column** is exactly balanced by the atmospheric pressure pushing down on the mercury in the trough. It works on the principle of **hydrostatic equilibrium**.

What the height represents 1 mk: The **height h of the mercury column** directly represents atmospheric pressure. At sea level, $h \approx 760$ mm Hg = 76 cm = 1 atm = 101.325 kPa.

OR

Pressure decreases with altitude 1.5 mk: As altitude increases, there is **less air column above**, so the weight of air pressing down is less. Therefore atmospheric pressure decreases. At very high altitudes (e.g., Mt Everest), pressure is only about 1/3 of sea-level pressure.

Cooking at high altitude 1.5 mk: Lower atmospheric pressure means water boils at a **lower temperature** (less than 100°C). Food therefore cooks at a lower temperature, requiring much longer cooking times. Turning up the heat does not help — water still boils at the lower temperature.

2(vii) Liquid Pressure P = ρgh 3 marks

Step 1 1 mk: Consider a liquid column of height h, base area A, density ρ. Mass $m = \rho V = \rho A h$. Weight/Force: $F = W = mg = \rho A h g$.

Step 2 1 mk: Pressure = Force/Area: `P = F/A = ρAhg/A`

Result 1 mk: `P = ρgh` — liquid pressure depends on (1) **density ρ** of the liquid and (2) **depth h** below the surface.

OR

Given: h = 500 m; ρ = 1025 kg/m³; g = 9.8 m/s²

Formula 1 mk: $P = \rho g h$

Substitution 1 mk: $P = 1025 \times 9.8 \times 500$

Answer 1 mk: $P = 5{,}022{,}500 \text{ Pa} \approx 5.02 \times 10^6 \text{ Pa} = 5.02 \text{ MPa}$

2(viii) Manometer 3 marks

Definition 1 mk: A **manometer** is a device used to measure the pressure in a fluid (gas or liquid) using fluid dynamics. It measures pressure relative to atmospheric pressure.

Working principle 1 mk: One end of the manometer is connected to the gas/fluid source; the other end is open to the atmosphere. If the source pressure exceeds 1 atm, the fluid is forced down on the source side and rises on the open side. The pressure is read from the height difference h.

Formula 1 mk: `P = ρgh`, where ρ = density of manometer fluid, g = 9.8 m/s², h = height difference of fluid columns.

OR

Four applications of a manometer 3 mk (0.75 each):

- Measuring the **pressure of fluids** in pipes and vessels using their mechanical properties.
- Measuring **vacuum** in sealed systems.
- Measuring the **flow rate** of fluids in pipelines (differential pressure manometer).
- Measuring the **filter pressure drop** in industrial filtration systems.

Also used for: meter calibrations, leak testing, and measuring liquid levels in tanks.

2(ix) Pascal's Principle **3 marks**

Pascal's Principle **1 mk**: "**An external pressure applied to an enclosed fluid is transmitted unchanged to every point within the fluid.**" (Blaise Pascal, 1623–1662)

Bicycle tyre example **2 mk**: When we pump a bicycle tyre, the pump applies force on the air inside. The air responds by pushing *not only against the pump piston* but also *equally against all walls of the tyre*. The pressure increases by the **same amount throughout the entire tyre** — this is Pascal's Principle at work. If you pump 10 Pa extra, every point inside the tyre increases by 10 Pa.

OR

Feature	Manometer	Barometer
Purpose	Measures pressure of a fluid (gas/liquid) relative to atmosphere **1 mk**	Measures **atmospheric pressure** of air **1 mk**
Design	Comes in different forms (U-tube, well-reservoir, inclined)	One basic design for all types
Fluid used	Mercury or any heavy liquid (sometimes lighter liquid)	Always mercury or heavy liquid only **1 mk**

2(x) Hydraulic Lift **3 marks**

Working **1 mk**: A hydraulic lift has two cylinders of different cross-sectional areas A_1 (small) and A_2 (large), connected by a tube filled with hydraulic fluid. A small input force F_1 is applied to piston 1.

Equation derivation **1 mk**: By Pascal's Principle, $P_1 = P_2$. Therefore: $F_1/A_1 = F_2/A_2$, giving

$F_2 = (A_2/A_1) \times F_1$

Force multiplier **1 mk**: Since $A_2 > A_1$, $F_2 > F_1$. A small force on the small piston creates a much larger force on the large piston. This is the **force multiplier** effect — hydraulic lifts can raise cars with the effort of a hand pump.

OR

Hydraulic Car Brake System (step by step) **3 mk**:

```
Brake Pedal pressed
       |
       v
Master Cylinder (piston increases
pressure in brake fluid)
       |
       v
Brake Lines (pressure transmitted
equally to all 4 wheels via Pascal's
Principle — brake fluid is enclosed)
       |
       v
Brake Calipers / Wheel Cylinders
(force applied to brake pads)
       |
       v
Brake Pads press on Disk Rotor
--> Friction slows the wheel
```

2(xi) Static Fluids Exert Perpendicular Force `3 marks`

Reason — Pascal's Principle `2 mk`: In a static fluid, pressure is transmitted **equally in all directions**. The force generated by pressure is F = P × A, acting perpendicular to any surface area A. If there were any component of force *parallel* to the surface, by Newton's third law the fluid would exert a force on the object parallel to the surface — this would cause uneven distribution of forces and contradicts the uniform nature of Pascal's Principle. Therefore static fluids can only exert forces **at right angles (perpendicular) to any surface**.

Example `1 mk`: A diver underwater feels pressure acting inward on every surface of their body, always perpendicular to that surface — on the chest (inward horizontally), on the top of the head (downward), on the soles of the feet (upward). This is why divers feel uniform "squeezing" rather than a force in one direction.

OR

Why divers wear special suits `2 mk`: At great depth, liquid pressure P = ρgh is enormous. Divers need suits that can withstand this **high inward pressure** on all body surfaces. Without protection, the pressure would crush the diver's body, rupture blood vessels, and collapse the lungs.

At 8.5 km depth `1 mk`: P = ρgh = 1000 × 9.8 × 8500 = **8.33×10^7 Pa = 83.3 MPa** — about 823 times atmospheric pressure. No human body could survive this without a rigid pressure suit.

SECTION C — Long Questions (4 × 5 = 20 Marks)

3(i) Elasticity, Elastic Limit, Hooke's Law & Force-Extension Graph `5 marks`

Elasticity `1 mk`: The ability of a deformed body to return to its original shape and size when the deforming force is removed. Materials exhibiting this are called **elastic materials** (e.g., spring, rubber, metals within limit).

Elastic Limit `1 mk`: The maximum force beyond which the material will not return to its original dimensions after the force is removed. Beyond the elastic limit, **permanent (plastic) deformation** occurs.

Hooke's Law `1 mk`: "Within the elastic limits, the extension or compression of a spring is directly proportional to the applied force." Mathematical form: $F_{res} = -kx$ ⇒ $k = F/x$, where k = **spring constant** (N/m), x = extension (m).

Force-Extension Graph `2 mk`:

```
Force (F) /N
     |                      * D (Breaking point)
     |                   *
     |               * C (Ultimate stress – max force)
     |           *
     |       * B (Elastic limit / Yield strength)
     |   * A (Proportional limit)
     | /  <-- straight line, gradient = k
     |/
     +-------------------------------------> Extension (x) /m
     O

Elastic region: O to B  |  Plastic region: B to D
```

Gradient of straight section OA = spring constant k. Beyond A: F no longer ∝ x. Beyond B: permanent deformation. At D: material breaks (fractures).

OR

Stress-Strain Curve for a Metal `5 marks`:

```
Stress (σ) /Pa
     |                     C (Ultimate stress)
     |                  *     *
     |               *           * D (Breaking/fracture point)
     |           *
     |       * B (Elastic limit / Yield strength)
     |   * A (Proportional limit)
     | /
     |/
     +-------------------------------------> Strain (ε) (dimensionless)
     O

 |---Elastic region---|-------Plastic region--------|
     O to B                    B to D
```

A — Proportional Limit `1 mk`: Up to A, stress ∝ strain (Hooke's Law obeyed). Graph is a straight line through origin.

B — Elastic Limit (Yield Strength) `1 mk`: Beyond A but up to B, material is still elastic (returns to original shape) but Hooke's Law is no longer obeyed.

C — Ultimate Stress `1 mk`: Maximum stress the material can bear before it begins to neck and thin. Any additional stress causes rapid failure.

D — Breaking Point `1 mk`: The material fractures (breaks) at D. Actual fracture stress is less than ultimate stress because of necking.

Elastic vs Plastic region `1 mk`: In the elastic region (O→B), material returns to original shape on unloading. In the plastic region (B→D), permanent deformation remains after unloading.

3(ii) Pressure: Definition, Area Effect, Real-Life Examples + Calculation 5 marks

Definition 1 mk: **Pressure** is defined as the force applied per unit area. Formula: `P = F/A`. SI unit: **pascal (Pa)** = 1 N/m². Pressure increases when force increases OR when area decreases.

Effect of area on pressure 2 mk (examples):

- **Thumb pin (drawing pin)**: The sharp tip has a tiny area, so even a small pressing force creates enormous pressure, allowing it to penetrate surfaces easily.
- **High-heel shoes**: A person's weight concentrated on the small heel area creates very high pressure on the floor — heels can damage soft flooring. Flat shoes spread the same weight over a larger area, reducing pressure.
- **Sharp knife**: The thin cutting edge has minimal contact area, so the same cutting force generates high pressure, slicing through food easily. A blunt knife has more area, lower pressure, and cuts poorly.

Calculation 2 mk:

Given: m = 50 kg; A = 2 cm² = 2×10^{-4} m²; g = 9.8 m/s²

Weight F = mg = 50 × 9.8 = 490 N

$$P = F/A = 490 / (2 \times 10^{-4}) = 2.45 \times 10^{6} \text{ Pa} = 2.45 \text{ MPa}$$

This enormous pressure (24,150 times atmospheric pressure on that tiny heel!) explains why pencil-heel shoes damage wooden floors.

OR

Atmospheric Pressure — full explanation 5 marks:

Definition 1 mk: "The pressure that atmospheric particles exert on the surface of the earth and all objects on it is called **atmospheric pressure**." At sea level: **1.013×10^5 Pa = 1 atm = 760 mmHg = 101.325 kPa**. 1 bar = 1.000×10^5 Pa.

Mercury Barometer — construction 2 mk: A glass tube (~80 cm long, closed at one end) is completely filled with mercury and inverted into a mercury trough. Mercury falls until the weight of the column balances atmospheric pressure. A vacuum forms at the closed top end. At sea level, the column height h ≈ **760 mm Hg** = 76 cm.

Variation with altitude 1 mk: As altitude increases, the weight of air column above decreases, so atmospheric pressure decreases. 1 atm = 760 mmHg at sea level; much less at high altitudes.

Weather indication 1 mk: Falling barometer reading indicates approaching low pressure (storms, rain). Rising reading indicates high pressure (clear, fair weather). Weather maps use **isobars** (lines of equal pressure) in millibars (1 mbar = 100 Pa).

3(iii) Liquid Pressure Derivation P = ρgh + Submarine Calculation **5 marks**

Derivation of P = ρgh **3 mk**:

Consider a cylindrical liquid column of:

- Height: h
- Base area: A
- Liquid density: ρ

Step 1: Volume of cylinder $V = Ah$ **1 mk**

Step 2: Mass $m = \rho V = \rho Ah$; Weight (Force) $F = W = mg = \rho Ahg$ **1 mk**

Step 3: Pressure $P = F/A = \rho Ahg/A \Rightarrow$ $P = \rho gh$ **1 mk**

Liquid pressure depends only on **density ρ** and **depth h** — not on the shape or total volume of the container.

Drill-hole activity: Water shoots fastest and furthest from the lowest hole because pressure is greatest at greater depth ($P = \rho gh$; larger $h \Rightarrow$ larger P).

Submarine Calculation **2 mk**:

Given: $h = 8.5$ km $= 8500$ m; $\rho = 1000$ kg/m^3; $g = 9.8$ m/s^2

$$P = \rho gh = 1000 \times 9.8 \times 8500 = 8.33 \times 10^7 \text{ Pa} = 83.3 \text{ MPa}$$

This is about 823 times atmospheric pressure — submarines must have thick reinforced hulls to withstand it.

OR

Manometer **5 marks**:

Definition **1 mk**: A manometer is a device used to measure pressure in a fluid using fluid dynamics.

Formula: $P = \rho gh$.

Three types (with diagrams described) **2 mk**:

```
   (a) U-tube Manometer        (b) Well-Reservoir           (c) Inclined Manometer
    Low   High                     High                         High
     |     |                        |                             |
     |     |                   [Reservoir]                    (angled tube,
     |_____|                        |                        gives better
     (U-shaped tube                 |                         resolution at
      with fluid                   tube                      low pressures)
      level shows                   up
      pressure diff)
```

Five applications **2 mk**:

- Measuring fluid pressure using mechanical properties of fluids
- Measuring vacuum in systems
- Measuring fluid flow rate
- Measuring filter pressure drop
- Meter calibrations and leak testing

Manometer vs Barometer: Barometer measures *atmospheric* pressure; manometer measures pressure of a fluid *relative to* atmospheric pressure.

3(iv) Pascal's Principle, Hydraulic Lift Derivation + Calculation 5 marks

Pascal's Principle 1 mk: "**An external pressure applied to an enclosed fluid is transmitted unchanged to every point within the fluid.**" The force generated by this pressure acts *perpendicular* to every surface it contacts.

Hydraulic Lift Derivation 2 mk:

Two connected cylinders with pistons:

- Piston 1: area A_1 (small), input force F_1
- Piston 2: area A_2 (large, $A_2 > A_1$), output force F_2

Pressure at piston 1: $P_1 = F_1/A_1$

Pressure at piston 2: $P_2 = F_2/A_2$

By Pascal's Principle: $P_1 = P_2$

$$F_1/A_1 = F_2/A_2 \Rightarrow F_2 = (A_2/A_1) \times F_1$$

Force multiplier: Since $A_2/A_1 > 1$, $F_2 > F_1$. A small effort on piston 1 lifts a heavy load on piston 2.

Calculation 2 mk:

Given: $A_1 = 0.002$ m²; $A_2 = 0.9$ m²; m = 1800 kg; g = 9.8 m/s²

$F_2 = W = mg = 1800 \times 9.8 = 17{,}640$ N

$$F_1 = (A_1/A_2) \times F_2 = (0.002/0.9) \times 17{,}640 = \mathbf{39.2\ N}$$

A force of only 39.2 N (about 4 kg weight) is sufficient to lift an 1800 kg car — demonstrating the enormous **mechanical advantage** of a hydraulic system.

OR

Hydraulic Car Brake System 5 marks:

```
HYDRAULIC CAR BRAKE SYSTEM
================================
1. BRAKE PEDAL (Force Input)
   Driver presses pedal
   --> Exerts force on master cylinder piston
   [1 mk]

2. MASTER CYLINDER (Pressure Increase)
   Piston compresses brake fluid
   --> Increases pressure throughout the fluid
   [1 mk]

3. BRAKE LINES (Pressure Transmission)
   By Pascal's Principle: increased pressure
   transmitted EQUALLY and UNCHANGED through
   all brake fluid to all four wheels
   simultaneously
   [1 mk]

4. BRAKE CALIPERS / WHEEL CYLINDERS (Force Application)
   Pressure acts on caliper pistons at each wheel
   --> Pistons push brake pads against disk rotor
   --> Friction slows and stops the wheel
   [1 mk]
```

Why Pascal's Principle ensures equal braking 1 mk: Because pressure is transmitted *unchanged* to all points in the enclosed brake fluid, *equal pressure* reaches all four wheel cylinders simultaneously. Equal pressure on equal-area caliper pistons produces equal force on all brake pads — the car brakes evenly without pulling to one side.

Air in brake fluid: Air is compressible (unlike fluid); if air enters the lines, the pedal "goes spongy" — the air compresses instead of transmitting pressure, and brakes fail. This is a serious safety hazard.

Chapters Covered: Unit 5 — Pressure and Deformation in Solids

— End of Answer Key —